

Mikhail Zyuzin

Senior Software Engineer • Data Architect • PostgreSQL • AWS Redshift • ServiceNow
Broadlands, VA • (703) 728-6106 • mvzyuzin@yahoo.com • [Mikhail Zyuzin LinkedIn Profile](#)

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Summary

Mikhail Zyuzin is a Senior Software Engineer and Data Architect with 20+ years of experience designing high-performance data platforms, analytics, and automation solutions for federal agencies including IRS, FAA, SEC, CBP, and DHS. Expert in SQL optimization, distributed systems, AWS Redshift, PostgreSQL, and large-scale data engineering. Proven results in reducing processing times from hours to minutes and delivering scalable enterprise automation solutions.

Career Highlights

- **Reduced IRS taxpayer conversion processing from 24 hours to 5 minutes**
- **Developed unique mathematical algorithms parsing all 216 IRS tax transmission forms**
- **Reduced SEC nanosecond precision trade processing from 2 hours to 10 seconds**
- **Accelerated CBP ADIS processing from 40 hours to under 10 minutes**
- **Improved FAA Jira reindexing from 3 hours to under 5 minutes**

Skills

PostgreSQL, Oracle, Greenplum, AWS Redshift, Java, Python, SQL, ServiceNow, Jira, Linux, AWS

Certifications

AWS Certified Cloud Practitioner (2025)

Experience

Zyuzin Technologies (2026-Present)

Senior Software Engineer / Data Architect

PostgreSQL, Java, JavaScript, Jira, ServiceNow

- Designed and developed an online Sudoku Generator and Solver using JavaScript and mathematical solving algorithms.

[Zyuzin Sudoku – Online Sudoku Generator and Solver](#)

Internal Revenue Service (2022–2025)

Senior Software Engineer / Data Architect

PostgreSQL, Greenplum, Oracle, AWS Redshift, Python, Java, Linux, Jira, ServiceNow

- Automated Greenplum backup, restore, and maintenance for all IRS Development, Test, and Production clusters.
- Resolved complex IRS MEF, Thomson Reuters, and FYT data mapping issues impacting production.
- Developed mathematical algorithms parsing all 216 IRS tax transmission forms.
- Detected multi-million duplicate records in CFTC Redshift production clusters.
- Designed petabyte-scale AWS Redshift composite key algorithms.

► View additional accomplishments

General Dynamics Information Technology (2018-2022)

Senior Data Engineer / Jira Architect

Jira, PostgreSQL, Oracle, AWS Redshift, Java, Python

- Maintained Transformation Data Management system for over 40 Federal Aviation Administration organizations.
- Supported and improved Department of Transportation Jira architecture.
- Accelerated Jira background reindexing from 3 hours to under 5 minutes.
- Created 200+ dashboards and dynamic Jira reports.
- Resolved over 3,000 production tasks and defects, significantly improving FAA applications.

► View additional accomplishments

U.S. Securities and Exchange Commission (2016-2018)

Senior Data Architect

Netezza, PostgreSQL, AWS Redshift, Java, C++, Linux, Jira

- Developed 250+ Netezza Trade and Quote analytical functions accelerating SEC applications.
- Created Netezza User Defined Function C++ library accelerating trillion-record database operations.
- Designed algorithms to identify prevailing quotes matching trades with nanosecond precision.
- Reduced multibillion-trade NBBO transaction processing from 2 hours to 10 seconds.

U.S. Customs and Border Protection (2016-2017)

Data Architect

Oracle, PostgreSQL, Java, Jira

- Remodeled CBP ADIS applications storing 15+ years of nationwide U.S. travel data.
- Developed algorithms for loading tables with referential constraints using optimized bulk collect and hint techniques, improving performance over existing DHS implementations.
- Reduced CBP ADIS application processing time from over 40 hours to under 10 minutes.

Internal Revenue Service (2013-2016)

Senior Software Engineer

PostgreSQL, Greenplum, Oracle, DB2, Java, Linux

- Developed algorithms converting over 2000 Oracle to Greenplum EFDS-RRP tables.
- Identified inefficient data swap manipulations causing excessive disk space consumption, enormous SQL updates and IRS production cluster collapse.
- Reduced IRS 1040 US nationwide taxpayer conversion time from 24 hours to 5 minutes.

► View additional accomplishments

Lockheed Martin (2005-2012)

Senior Software Engineer

Oracle, DB2, SQL, Java, C#

- Led software development team generating large-scale CAMIR, ANSI, BAPLIE, and CATAIR messages.
- Optimized IBM MQ server performance accelerating 1,000,000 element EDI processing from 10 hours to under 5 minutes.
- Resolved over 300 CBP production defects.

► View additional accomplishments

Education

PhD, Mathematics and Computer Science
Novosibirsk State University

MS, Mathematics and Computer Science
Novosibirsk State University

Awards

- Winner, Siberian Mathematics Olympiad
- USSR Exhibition of Industrial Achievements Medal
- First Place, Siberian Academy Scientific Competition (4-time recipient)
- Academician Marchuk Award